

⚠ Important

- Use Kiro CLI for this section
- Complete the [Setting Up Kiro](#) section

Kiro CLI - Custom Agents

Introduction

Custom agents in Kiro CLI are **customizable AI assistants** that you can configure for specific use cases and workflows. According to AWS documentation, instead of using a generic assistant that requires extensive context and prompt engineering, custom agents allow you to pre-configure the right set of tools, permissions, and context for different scenarios.

As stated in the [AWS announcement](#), custom agents "enable you to customize the CLI agent to perform specialized tasks such as code reviews and troubleshooting more effectively. You can define a custom agent by creating a configuration file that specifies which tools an agent is allowed to use, prompts to guide its behavior, and the context needed to perform the task." After the [migration to Kiro CLI](#), this feature remains fully available in Kiro CLI.

Core Architecture

Each custom agent is defined by a **JSON configuration file** that serves as a declarative specification for:

- **Tool Access:** Which built-in and MCP (Model Context Protocol) tools are available
- **Security Controls:** Granular permissions and access restrictions
- **Context Integration:** Static files and dynamic context hooks
- **Behavioral Guidelines:** Prompts and instructions that guide agent responses
- **Resource Management:** Project-specific or global configurations

How Custom Agents Differ from Default Experience

By default, Kiro CLI provides access to all available tools but requires user confirmation for most operations. This approach prioritizes security but can interrupt workflow with frequent permission prompts. Custom agents solve this by allowing you to:

- **Pre-approve specific tools** - Define which tools can run without prompting
- **Limit tool access** - Restrict which tools are available to reduce complexity
- **Include relevant context** - Automatically load project files, documentation, or system information
- **Configure tool behavior** - Set specific parameters for how tools should operate

Key Benefits

- **Workflow Optimization** - Create custom agents tailored to specific tasks like AWS infrastructure management, code reviews, or debugging sessions.
- **Reduced Interruptions** - Pre-approve trusted tools to eliminate permission prompts during focused work sessions.
- **Enhanced Context** - Automatically include relevant project documentation, configuration files, or system information.
- **Team Collaboration** - Share custom agent configurations with team members to ensure consistent development environments.
- **Security Control** - Limit tool access to only what's needed for specific workflows, reducing potential security risks.

📌 Custom agents represent a paradigm shift in developer tooling, transforming Kiro CLI from a generic AI assistant into a specialized, secure, and highly configurable platform for professional software development and operations.

This approach reduces context switching, streamlines the development experience, and provides developers with more granular control over AI-assisted workflows while maintaining the security and reliability expected in enterprise environments.

What You'll Learn

Throughout this lab, you'll discover how to:

- Understand custom agents architecture
- Create your first custom agent
- Master basic agent concepts
- Master the 5 core agent configuration patterns
- Build progressively complex agents

By the end of this lab, you'll be able to transform Kiro from a general-purpose code generator into a specialized development assistant that follows your preferred coding practices and architectural approaches.

Prerequisites

Before starting this lab, ensure you have:

- Completed the [Setting Up Kiro](#) section
- A basic understanding of web development concepts
- Familiarity with command-line interfaces

Let's begin! 🤖 ⚡